Ex no: 1

Date: 16/7/26

Basic Networking commands

AIM: To execute the basic networking commands in the windows and linux environment

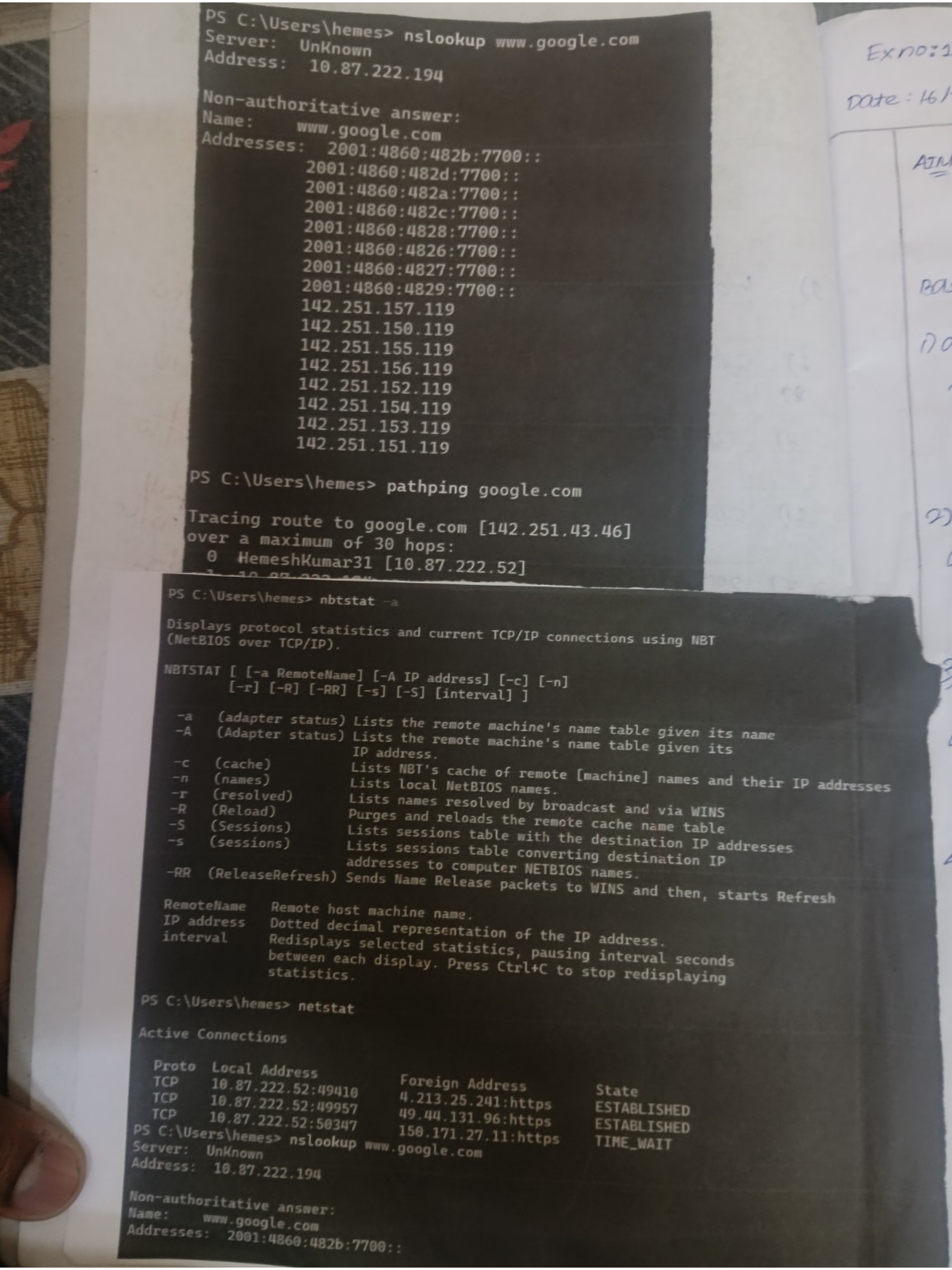
Basic commands:

1) arp-a : ARP is short form of address resolution protocal. It will show the IP computer along with IP address.

2) Hostname: This is the simplest of all TCP/IP commands. It display the computer's hostname (device name)

3) ipconfig/all : Display complete network configuration, including router, gateway DNS DHCP and type of ethernet

4) netstat: Display active network connections listening port. router table and network



```
PS C:\Users\hemes> nslookup www.google.com
Server:  UnKnown
Address:  10.87.222.194

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482b:7700::
          2001:4860:482d:7700::
          2001:4860:482a:7700::
          2001:4860:482c:7700::
          2001:4860:4828:7700::
          2001:4860:4826:7700::
          2001:4860:4827:7700::
          2001:4860:4829:7700::
          142.251.157.119
          142.251.150.119
          142.251.155.119
          142.251.156.119
          142.251.152.119
          142.251.154.119
          142.251.153.119
          142.251.151.119

PS C:\Users\hemes> pathping google.com

Tracing route to google.com [142.251.43.46]
over a maximum of 30 hops:
  0  HemeshKumar31 [10.87.222.52]
```

```
PS C:\Users\hemes> nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

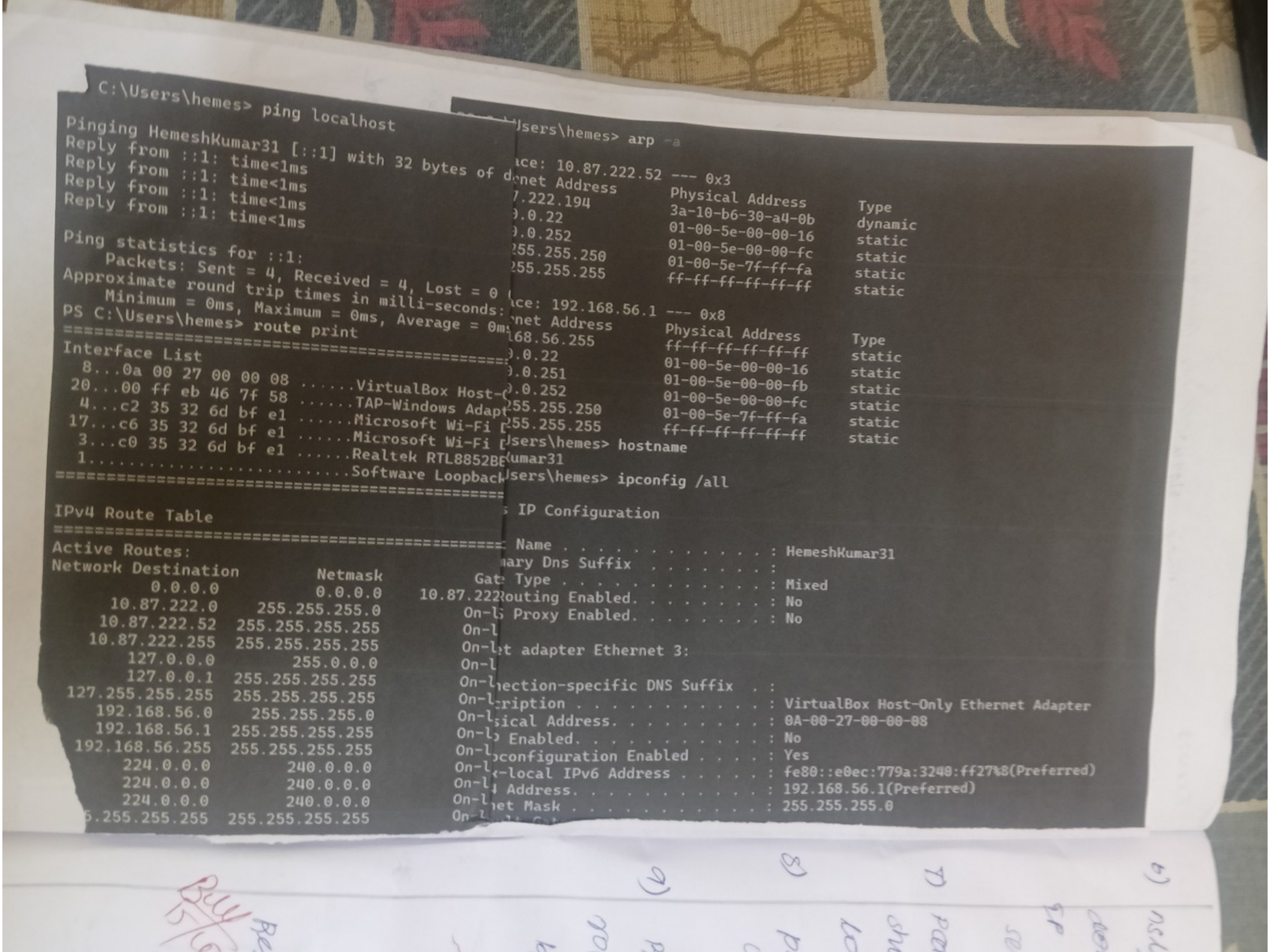
  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.

PS C:\Users\hemes> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.87.222.52:49410     4.213.25.241:https     ESTABLISHED
  TCP    10.87.222.52:49957     49.44.131.96:https     ESTABLISHED
  TCP    10.87.222.52:50347     150.171.27.11:https    TIME_WAIT
PS C:\Users\hemes> nslookup www.google.com
Server:  UnKnown
Address:  10.87.222.194

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482b:7700::
```



```
C:\Users\hemes> ping localhost

Pinging HemeshKumar31 [::1] with 32 bytes of d
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\hemes> route print
===========================================
Interface List
  8...0a 00 27 00 00 08 ......VirtualBox Host-
 20...00 ff eb 46 7f 58 ......TAP-Windows Adapt
  4...c2 35 32 6d bf e1 ......Microsoft Wi-Fi
 17...c6 35 32 6d bf e1 ......Microsoft Wi-Fi
  3...c0 35 32 6d bf e1 ......Realtek RTL8852BE
  1...........................Software Loopback
===========================================

IPv4 Route Table
===========================================
Active Routes:
Network Destination        Netmask          Gat
          0.0.0.0          0.0.0.0   10.87.222
       10.87.222.0    255.255.255.0         On-l
      10.87.222.52  255.255.255.255         On-l
     10.87.222.255  255.255.255.255         On-l
        127.0.0.0        255.0.0.0         On-l
        127.0.0.1  255.255.255.255         On-l
  127.255.255.255  255.255.255.255         On-l
     192.168.56.0    255.255.255.0         On-l
     192.168.56.1  255.255.255.255         On-l
   192.168.56.255  255.255.255.255         On-l
        224.0.0.0        240.0.0.0         On-l
        224.0.0.0        240.0.0.0         On-l
        224.0.0.0        240.0.0.0         On-l
  255.255.255.255  255.255.255.255         On-l
```

```
Users\hemes> arp -a

ace: 10.87.222.52 --- 0x3
net Address          Physical Address      Type
.222.194             3a-10-b6-30-a4-0b     dynamic
.0.22                01-00-5e-00-00-16     static
.0.252               01-00-5e-00-00-fc     static
255.255.250          01-00-5e-7f-ff-fa     static
255.255.255          ff-ff-ff-ff-ff-ff     static

ace: 192.168.56.1 --- 0x8
net Address          Physical Address      Type
168.56.255           ff-ff-ff-ff-ff-ff     static
.0.22                01-00-5e-00-00-16     static
.0.251               01-00-5e-00-00-fb     static
.0.252               01-00-5e-00-00-fc     static
255.255.250          01-00-5e-7f-ff-fa     static
255.255.255          ff-ff-ff-ff-ff-ff     static

Users\hemes> hostname
Kumar31

Users\hemes> ipconfig /all

 IP Configuration

   Name . . . . . . . . . . . . : HemeshKumar31
ary Dns Suffix  . . . . . . . :
   Type . . . . . . . . . . . . : Mixed
outing Enabled. . . . . . . . : No
   Proxy Enabled. . . . . . . . : No

t adapter Ethernet 3:

nection-specific DNS Suffix  . :
cription . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
sical Address. . . . . . . . . : 0A-00-27-00-00-08
   Enabled. . . . . . . . . . . : No
oconfiguration Enabled . . . . : Yes
k-local IPv6 Address . . . . . : fe80::e0ec:779a:3240:ff27%8(Preferred)
   Address. . . . . . . . . . . : 192.168.56.1(Preferred)
net Mask . . . . . . . . . . . : 255.255.255.0
```

6) nslookup: lookup DNS information for a domain or IP address. It return the IP address of the websites and DNS server details

7) pathping : combines ping and tracert shows the path to a destination and packet lose where packet loss occurs

8) ping: test wheather another device or website loss occurs

9) Route: Display or modifies the IP routes table shows all routes used by the computer to reach different network

Result:

Thus, the basic networking commands has been Successfully executed in the Linux.

BM 15/10